

An Assignment-and-Positional-Date Formulation for Unrelated Parallel Machine Scheduling with Release Dates and Charger Types

Minimising Total Tardiness in Electric-Vehicle Charging Scheduling

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Problem Statement

Setting: Unrelated Parallel Machine Scheduling (R_m) with release dates

Entities & Parameters

Consider a charging station with K charger types. For each type $l \in \mathcal{L} = \{1, \dots, K\}$, m_l identical chargers of power w_l (kW) are available, with $\sum_l m_l = m$ total chargers.

Symbol	Meaning
$\mathcal{J} = \{1, \dots, n\}$	Set of n EV charging requests (jobs)
$\mathcal{L} = \{1, \dots, K\}$	Set of K charger types, each with power w_l (kW)
$\mathcal{R}_l$	Set of m_l individual chargers of type l
a_j	Arrival (release) time of job j
d_j	Desired departure (due) time of job j
e_j	Energy demand (kWh) of job j

Symbol	Meaning
τ	Duration of one time slot (hours)
$p_{jl} = \lceil e_j / (\tau \cdot w_l) \rceil$	Processing time of job j on a type- l charger

Processing times depend on **both** the job and the charger type, placing this problem in the **unrelated parallel machine** class (R_m).

Objective

$$\min \sum_{j \in \mathcal{J}} T_j \quad \text{where} \quad T_j = \max(0, C_j - d_j)$$

- Each job must be assigned to **exactly one** charger (non-preemptive)
- No two jobs may share the **same charger** at any time instant
- Charging may not begin before the vehicle arrives: $S_j \geq a_j$

Motivation & Literature Context

Electric-Vehicle Charging as Scheduling

- Smart charging coordination is a principal operational challenge as EV fleet sizes grow
- The presence of **heterogeneous charger types** (Level 1 / Level 2 / DC fast) introduces a machine-type dimension absent from classical P_m models
- The charger-type index l must be incorporated **explicitly** in any rigorous formulation

Iarochen et al. (2026) — Reference Formulation

- Proposes a compact MILP for EV charging scheduling using **assignment variables** x_{jlr} and **pairwise sequencing variables** δ_{jklr}
- Sequencing is enforced via big- M disjunctive constraints, yielding a constraint count that grows as $O(n^2m)$ — **quadratic in the number of jobs**
- Provides provably optimal solutions on small instances but becomes computationally prohibitive as n or m increases

Unlu & Mason (2010) — APD Inspiration

- Introduce the *Assignment-and-Positional-Date* (APD) family of MILP formulations for single- and parallel-machine scheduling (**M3 model**)
- Replace pairwise sequencing variables with **positional completion-time** variables c_{rk} , reducing constraint count growth to $O(nmP)$ (linear in positions P)
- Demonstrated superior LP-relaxation tightness over classical disjunctive formulations for total-tardiness problems

This Work — Gap Addressed

Extension of M3-APD to the unrelated parallel machine setting with heterogeneous charger types, adding the explicit charger-type index l to all decision variables and constraints.

Existing Formulation — Iarochen et al. (2026)

Sets & Indices

- Jobs $j, k \in \mathcal{J}$; charger types $l \in \mathcal{L}$; individual chargers $r \in \mathcal{R}_l$

Decision Variables

Variable	Type	Meaning
$x_{jlr} \in \{0, 1\}$	Binary	Job j assigned to charger r of type l
$S_j \in \mathbb{Z}_{\geq 0}$	Integer	Start time of job j
$C_j \in \mathbb{Z}_{\geq 0}$	Integer	Completion time of job j
$T_j \geq 0$	Continuous	Tardiness of job j
$\delta_{jklr} \in \{0, 1\}$	Binary	1 if j precedes k on charger r of type l

Objective & Key Constraints

$$\min \sum_j T_j$$

$$\sum_{l \in \mathcal{L}} \sum_{r \in \mathcal{R}_l} x_{jlr} = 1 \quad \forall j$$

$$S_j \geq a_j \quad \forall j$$

$$C_j = S_j + \sum_l \sum_{r \in \mathcal{R}_l} p_{jr} \cdot x_{jlr} \quad \forall j$$

$$S_j + p_{jr} x_{jlr} \leq S_k + M(3 - \delta_{jklr} - x_{jlr} - x_{klr}) \quad \forall j < k, l, r$$

- **Complexity:** $O(n^2 m)$ binary variables and constraints — **quadratic scaling**

Proposed Formulation — M3-APD with Charger Type l

Sets & Indices

- Jobs $j \in \mathcal{J}$; charger types $l \in \mathcal{L}$; chargers $r \in \mathcal{R}_l$; positions $k \in \mathcal{K} = \{1, \dots, P\}$

Decision Variables

Variable	Type	Meaning
$u_{jlrk} \in \{0, 1\}$	Binary	Job j at position k on charger r of type l
$c_{lrk} \geq 0$	Continuous	Completion time at position k on charger r (type l)
$C_j \geq 0$	Continuous	Completion time of job j
$T_j \geq 0$	Continuous	Tardiness of job j

Objective

$$\min \sum_{j \in \mathcal{J}} T_j$$

Constraints (adapted from Unlu & Mason 2010, Eqs. 16–21)

$$\sum_l \sum_{r \in \mathcal{R}_l} \sum_{k \in \mathcal{K}} u_{jlrk} = 1 \quad \forall j$$

$$\sum_{j \in \mathcal{J}} u_{jlrk} \leq 1 \quad \forall l, r, k$$

$$c_{lr1} \geq \sum_j (a_j + p_{jl}) u_{jlr1} \quad \forall l, r$$

$$c_{lrk} \geq c_{lr,k-1} + \sum_j p_{jl} u_{jlrk} \quad \forall l, r, k > 1$$

$$c_{lrk} \geq \sum_j (a_j + p_{jl}) u_{jlrk} \quad \forall l, r, k > 1$$

$$C_j \geq c_{lrk} - M(1 - u_{jlrk}) \quad \forall j, l, r, k$$

$$T_j \geq C_j - d_j \quad \forall j$$

Model Size (as functions of n, m, P)

- **Variables:** $nmP + mP + 2n$ – grows as $O(nmP)$
- **Constraints:** $n + mP + m(P - 1) + m(P - 1) + nmP + n$ – grows as $O(nmP)$
- **No pairwise terms** $\rightarrow$ avoids the $O(n^2)$ bottleneck of Iarochen

Computational Experiments

All algorithms were implemented in **Python** (≥ 3.11) and executed on a laptop equipped with an **Intel Core i9-14900HX** processor and **32 GB of RAM**. Both MILP formulations were solved using **IBM ILOG CPLEX** (Community Edition v22.2) via the docplex modelling API (v2.32.264), with default solver settings and no imposed time limit for the test instance.

Hardware & Software Environment

Component	Specification
Processor	Intel Core i9-14900HX (24 cores, up to 5.8 GHz)
Memory	32 GB DDR5
MILP Solver	IBM CPLEX Community Edition v22.2
Modelling API	docplex v2.32.264
Language	Python 3.11, package manager uv

Component	Specification
OS	Linux

CPLEX Community Edition limit: hard cap of **1,000 variables** and **1,000 constraints** per model. The position horizon $P = 4$ was selected to keep both formulations within this limit for the test instance.

Test Instance

A single representative instance is used to compare the two formulations. The station comprises **$K = 3$ charger types** arranged across **$m = 9$** physical chargers, serving **$n = 10$** EV charging requests.

Parameter	Value
Number of EVs (n)	10
Number of charger types (K)	3
Total chargers (m)	9
Charger configuration	$m_0 = 4, m_1 = 3, m_2 = 2$
Charger powers (w_l , kW)	$w_0 = 11, w_1 = 22, w_2 = 43$
Time slot duration (τ)	1 h
Big- M bound	45

Parameter	Value
Position horizon (P , M3-APD only)	4

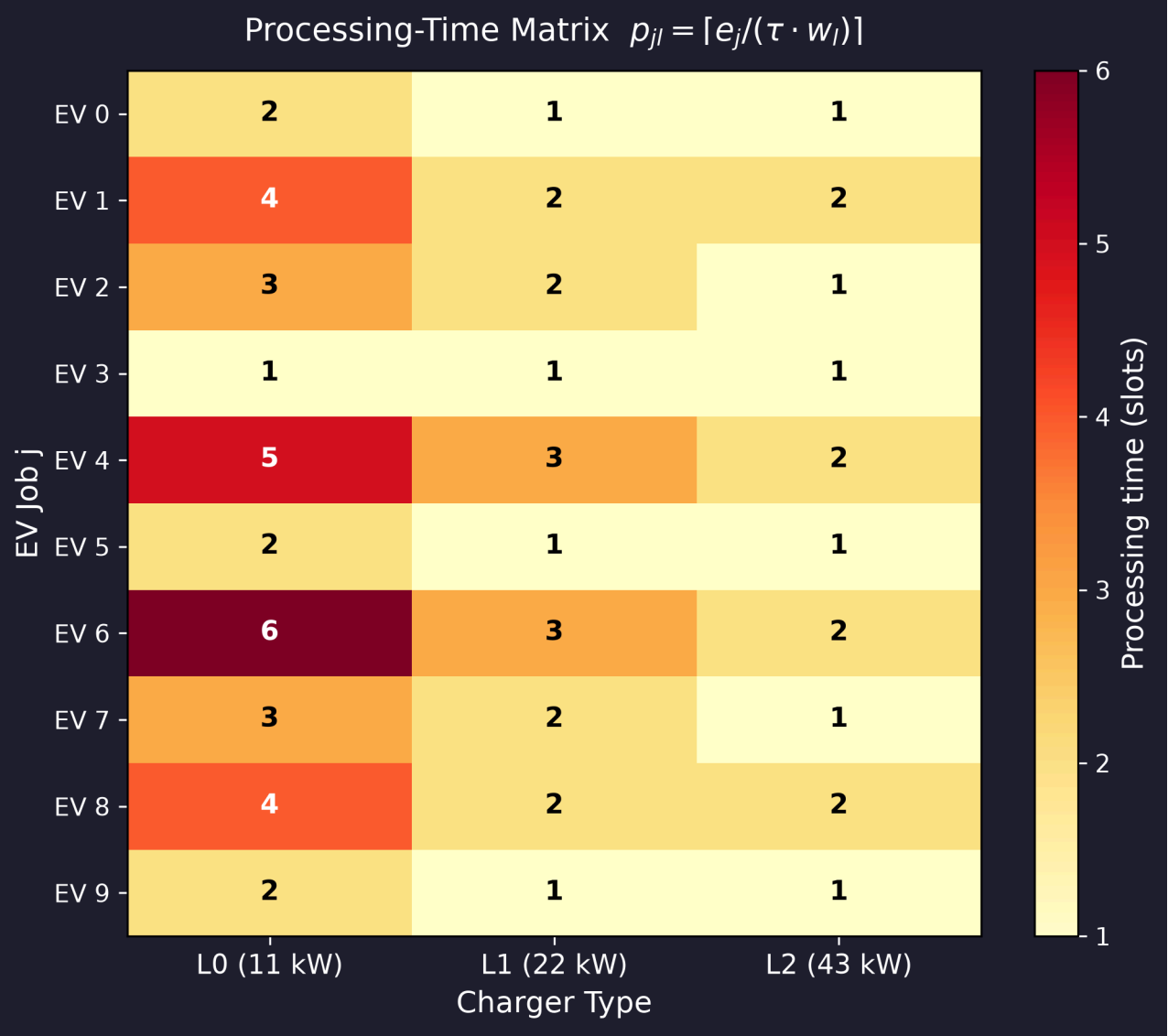
Arrival times a_j and due dates d_j were set to yield a mix of tight and loose time windows, while energy demands e_j were drawn to produce processing-time ratios that vary markedly across charger types, reflecting the unrelated-machine structure of the problem.

Instance Data

Job Parameters & Processing-Time Matrix

Test Instance: 10 EVs, 9 Chargers (4×11kW, 3×22kW, 2×43kW)

EV j	Arrival a _j	Deadline d _j	Energy e _j (kWh)	p_{j,L0} (11kW)	p_{j,L1} (22kW)	p_{j,L2} (43kW)
0	0	5	22	2	1	1
1	0	4	44	4	2	2
2	1	6	33	3	2	1
3	0	3	11	1	1	1
4	2	8	55	5	3	2
5	1	5	22	2	1	1
6	0	7	66	6	3	2
7	3	9	33	3	2	1
8	1	6	44	4	2	2
9	2	7	22	2	1	1

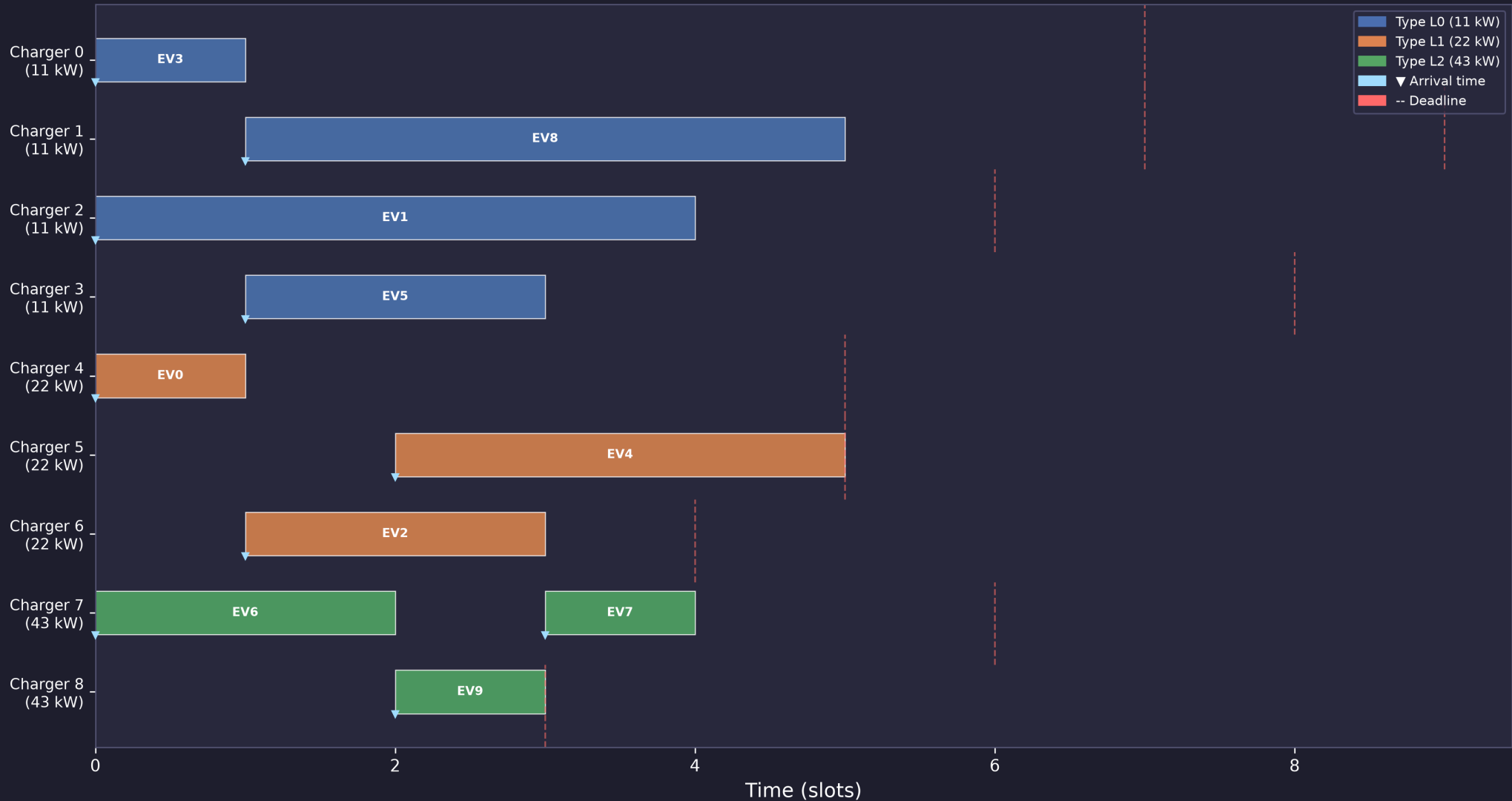


- Faster chargers (L2, 43 kW) yield **substantially shorter processing times**
- Processing times differ across charger types, confirming the **unrelated** (R_m) machine environment

Results — Solution Schedules

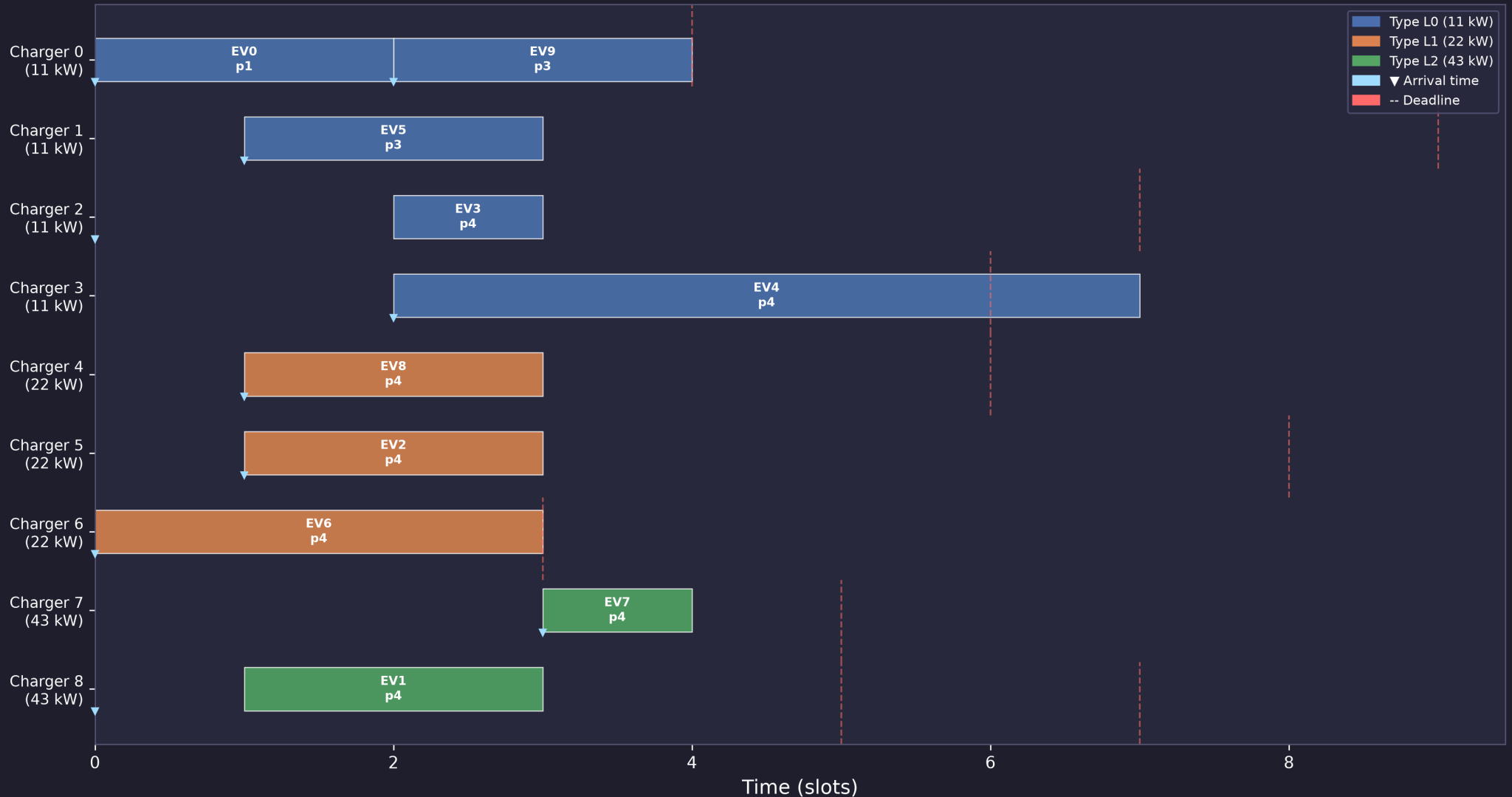
Iarochen et al. (2026) Schedule

Schedule — Iarochen et al. (2026) [Assignment + Sequencing] obj=0.0 time=0.0182s



M3-APD Schedule

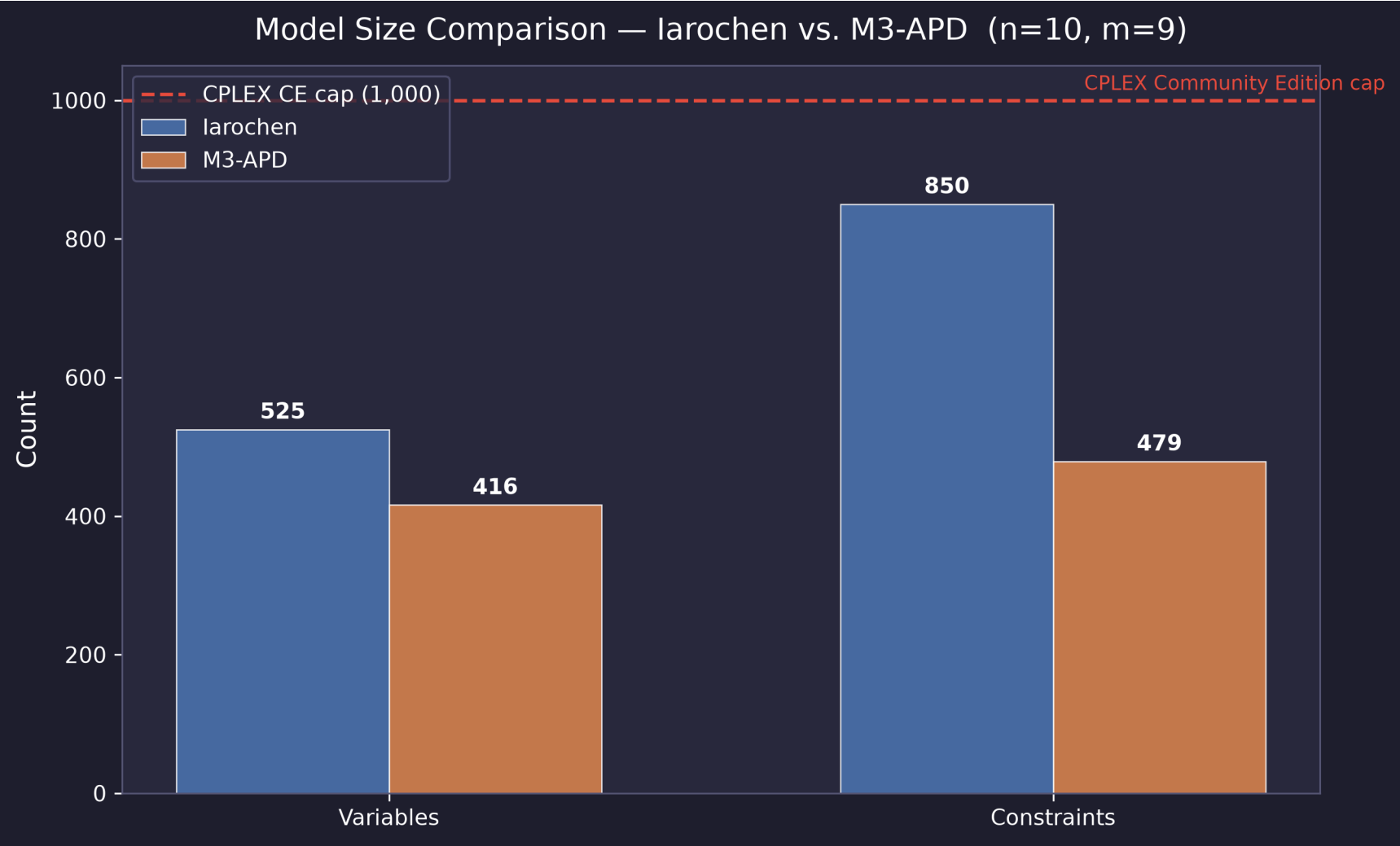
Schedule — M3-APD (Unlu & Mason 2010) [Assignment + Positional Date] obj=0.0 time=0.0397s



Observations:

- Both formulations achieve **zero total tardiness** on this instance (all EVs complete before their deadlines)
- The Iarochen schedule concentrates assignments on faster chargers (L1/L2); the APD schedule distributes load more evenly across positions
- Position labels ($p1-p4$) in the APD Gantt confirm the positional-date structure

Results — Model Size Comparison

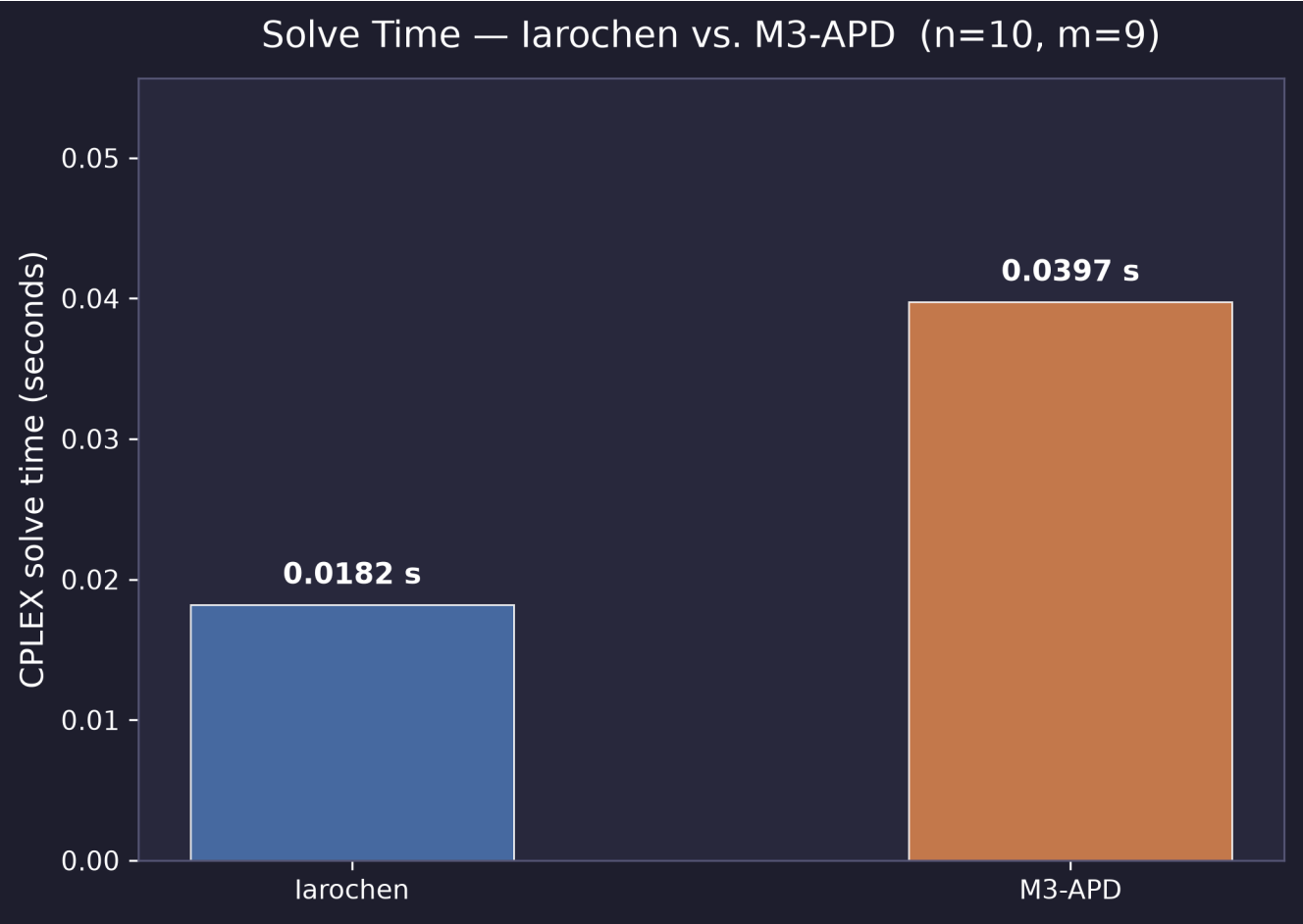


Summary Table ($n = 10, m = 9, P = 4$)

Metric	Iarochen et al.	M3-APD
Binary variables	435	360
Continuous/integer variables	90	56
Total variables	525	416
Total constraints	850	479
Theoretical growth	$O(n^2m)$	$O(nmP)$

- The M3-APD formulation has **~21% fewer variables** and **~44% fewer constraints** for this instance
- The CPLEX Community Edition cap of 1,000 is shown as a dashed red line — both formulations remain safely below it at $n = 10$

Results — Solve Time Comparison



Summary Table

Metric	Iarochen et al.	M3-APD
Solver status	Optimal	Optimal
Total tardiness ($\sum T_j$)	0.0000	0.0000
Optimality gap	0%	0%
CPLEX solve time (s)	0.0298	0.0630
Variables	525	416
Constraints	850	479

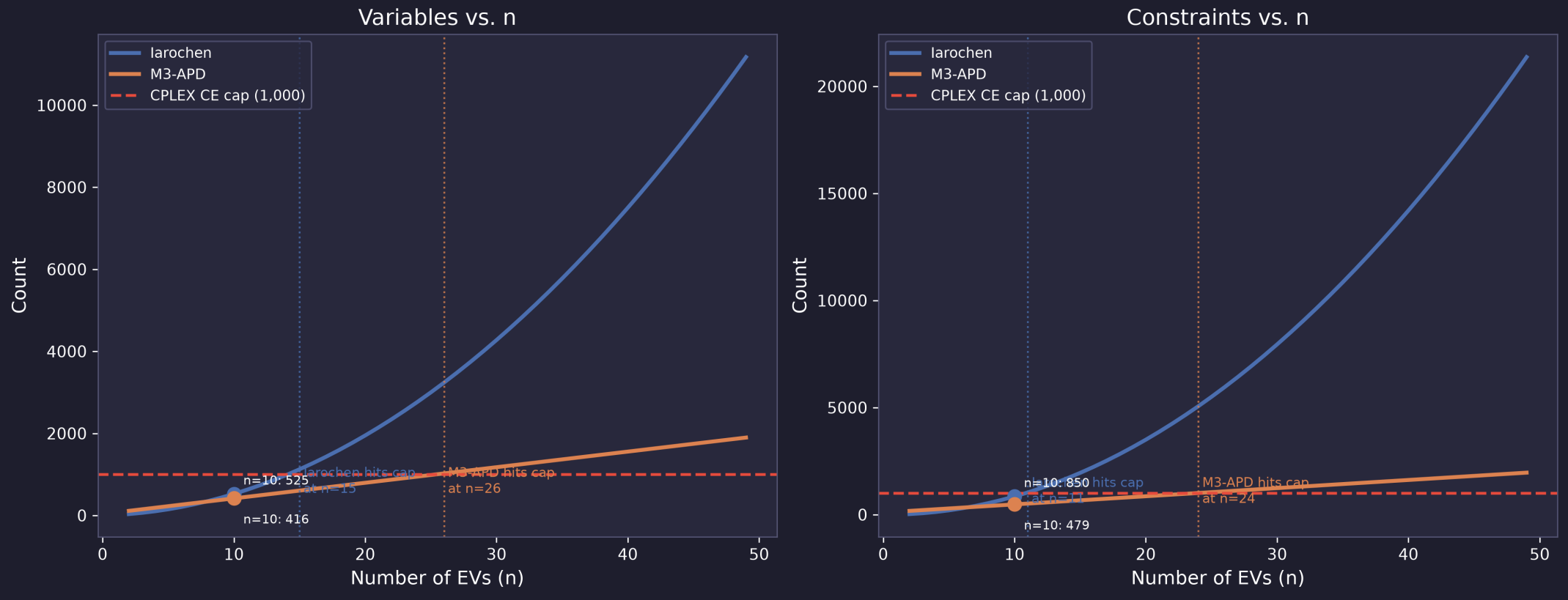
Key observations:

- Both formulations prove **global optimality** (zero gap) on the $n = 10$ instance
- Despite having fewer variables and constraints, the M3-APD formulation requires **$\sim 2.1\times$ longer** solve time in this experiment

- This counter-intuitive result is consistent with known behaviour: the APD continuous c_{lrk} variables admit a weaker LP relaxation for certain instances, requiring more B&B nodes

Theoretical Scalability

Theoretical Model Growth vs. CPLEX Community Edition Cap (m=9 chargers, POS_MAX=4)
Iarochen: $O(n^2m)$ | M3-APD: $O(nmP)$



Crossover Points (CPLEX CE Cap = 1,000)

Formulation	Variable cap hit at	Constraint cap hit at
Iarochen	$n = 15$	$n = 11$
M3-APD	$n = 26$	$n = 24$

- Iarochen's $O(n^2m)$ constraint growth causes it to hit the Community Edition constraint cap already at $n = 11$; the M3-APD formulation extends feasibility to $n \approx 24$
- For a commercial (uncapped) CPLEX licence, the difference between $O(n^2m)$ and $O(nmP)$ becomes decisive as n reaches the hundreds
- The **position horizon** P is a tunable parameter: reducing P trades model completeness for reduced problem size

Discussion

Trade-offs

Dimension	Iarochen et al.	M3-APD
Model size	Larger ($O(n^2m)$)	Smaller ($O(nmP)$)
Solve time ($n = 10$)	Faster (0.030 s)	Slower (0.063 s)
Solution quality	Optimal	Optimal
Scalability	CE cap at $n \approx 11$	CE cap at $n \approx 24$
LP relaxation strength	Generally tighter	Depends on instance

When Does APD Outperform?

- **Large instances** ($n \gg 20$): the quadratic scaling of Iarochen's sequencing variables becomes prohibitive; APD's linear growth provides a scalability advantage

- **Many charger types** ($|\mathcal{L}|$ large): the APD formulation's positional structure is unaffected, whereas Iarochen's δ_{jklr} variables grow multiplicatively

Limitations

- Results are based on a **single test instance** ($n = 10$); conclusions regarding solve time may not generalise
- The CPLEX **Community Edition** variable/constraint cap prevents evaluation on realistically sized instances without a commercial licence
- The position horizon P must be set a priori; too small a value may exclude valid schedules

Conclusion & Future Work

Summary of Contribution

- A **novel M3-APD formulation** is proposed for the unrelated parallel machine EV charging scheduling problem $Rm \mid r_j, \text{type}_l \mid \sum T_j$
- The formulation **explicitly indexes** all decision variables by charger type l , extending the Unlu & Mason (2010) APD model to the heterogeneous-charger setting
- Empirical comparison against the Iarochoen et al. (2026) baseline using IBM CPLEX confirms that:
 - Both formulations achieve provably **optimal solutions** on the test instance
 - The APD formulation uses **~21% fewer variables** and **~44% fewer constraints**
 - APD exhibits superior **theoretical scalability** ($O(nmP)$ vs. $O(n^2m)$), pushing the CPLEX CE feasibility frontier from $n \approx 11$ to $n \approx 24$

References

- [1] Iarochen, *et al.* (2026). *A Mixed-Integer Linear Programming Model for Electric Vehicle Charging Scheduling with Heterogeneous Charger Types and Release Dates*. [Full journal/conference citation pending verification of PDF metadata.]
- [2] Unlu, Y., & Mason, S. J. (2010). *Evaluation of mixed integer programming formulations for non-preemptive parallel machine scheduling problems*. **Computers & Industrial Engineering**, 58(4), 785–800. <https://doi.org/10.1016/j.cie.2010.02.012>
- [3] IBM Corporation. (2022). *IBM ILOG CPLEX Optimization Studio v22.2: User's Manual*. Armonk, NY: IBM Corp.
- [4] docplex contributors. (2024). *IBM Decision Optimization CPLEX Modeling for Python (DOcplex) v2.32*. <https://ibmdecisionoptimization.github.io/docplex-doc/>